

The Collatz and Erdős Problem 817 workbenches, and the bridges between the author's workbenches: an audited record

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Abstract

The author's research programme includes a Collatz workbench, a workbench on Erdős Problem 817, and workbenches on the Erdős–Straus equation, Yang–Mills theory, Navier–Stokes, the S^6 construction and the Riemann zeta function, with many explicit maps between them. For the Collatz problem we prove a sharp threshold for the full history law of consecutive starts (the word law of N starts is close to the geometric model below $(1/2)\log_2 N$ halving counts and far from it above), explicit infinite families of merging pairs, a descent criterion that is necessary and sufficient, a contraction in the fair-coin model, and cycle bounds: every nontrivial positive cycle has at least 1636 odd steps from the workbench's own verification range, at least 190,537 from a verification to 2^{33} made for this record, and at least 72,057,431,991 from Barina's published range by Eliahou's method. We also state an exact, invertible encoding of probability vectors on exponent words into the motion of a zero of the Riemann zeta function under the Hurwitz flow. For Problem 817 we give the proof, due to an anonymous contributor and reconstructed by the Clankers, that the least size $g_4(n)$ of a set whose subset sums avoid 4-term progressions satisfies $\lim g_4(n)^{1/n} = 19^{1/3}$, together with a variance refinement, a lifting principle, the characterization of every rate Λ_k as an infimum over modular certificates, the strict decrease $\Lambda_3 = 3 > \Lambda_4 = 19^{1/3} > 97^{1/6} \geq \Lambda_5$, certificates for $k = 5, 6$, and the exact values $g_4(1), \dots, g_4(6) = 1, 3, 5, 14, 40, 79$. Finally we describe the bridges between the workbenches as typed maps, with what each carries and what it suggests.

0. Introduction

0.1 Why this record

The workbenches were developed with AI systems over 2026, each with its own notes, formal checks and continuations, and they are connected by explicit maps: the same period system appears in the S^6 , Erdős–Straus and Yang–Mills workbenches, the Hurwitz flow of the zeta programme appears in the Collatz workbench, and identities recur across several of them. This record states what the Collatz and Problem 817 workbenches prove, with proofs, and describes the bridges between all the workbenches, so that a reader can see both the results and the structure that connects them.

0.2 Main results

Collatz (§1).

- **Theorem 1.2** (*proved here*). The total variation between the empirical law of the first m exponents over N consecutive odd starts and the geometric model has an explicit two-sided bound for every m ; it tends to $\Phi(2c)$ at $m = L/2 + c\sqrt{L}$ ($L = \log_2 N$), to 0 at a polynomial rate below $(1/2 - \delta)L$, and to 1 above $(1/2 + \delta)L$, uniformly in the offset.
- **Theorem 1.4**. Explicit arithmetic progressions of pairs $m < n$ whose trajectories meet, with prescribed exponent words.
- **Proposition 1.5** (*generalized here*). A cylinder of exponent words descends if and only if its least member does.
- **Theorem 1.8 and Corollary 1.9**. Every nontrivial positive cycle has at least 1636 odd steps and more than 678 steps with exponent 1 from the workbench's verification range; at least 190,537 odd steps from a verification to 2^{33} made here; and at least 72,057,431,991 odd steps from Barina's published range.
- **Theorem 1.10**. An exact encoding of weights on exponent words into the zero motion of the Hurwitz flow of ζ .

Erdős Problem 817 (§2).

- **Theorem 2.1.** $(19^{\lfloor n/3 \rfloor} - 1)/(2n) \leq g_4(n) \leq 8 \cdot 19^{\lfloor n/3 \rfloor - 1}$, so $\lim g_4(n)^{1/n} = 19^{1/3}$.
- **Proposition 2.5 and Theorem 2.7** (*generalized here*). A lifting principle, and the rates Λ_k as infima over modular certificates, with $\Lambda_3 = 3 > \Lambda_4 = 19^{1/3} > 97^{1/6} \geq \Lambda_5$.
- **Theorem 2.6.** A variance refinement of the lower bound.
- **Proposition 2.10** (*computed for this audit*). $g_4(1), \dots, g_4(6) = 1, 3, 5, 14, 40, 79$ and $176 \leq g_4(7) \leq 246$.

Bridges (§4). Twenty-eight typed edges between the workbenches, eight of them substantive; they are stated with what crosses each and what it suggests.

0.3 What the results mean

For the Collatz problem, the history threshold makes precise how far the classical cylinder structure of Terras and Everett reaches: below half the logarithm of the sample size the full word law of consecutive starts is geometric, above it the word law is far from geometric, and the transition is Gaussian. The cycle bounds show how the verified range of the conjecture turns into lower bounds on the length of any nontrivial cycle through Eliahou's argument, and they locate the workbench's own bound within the published ones. The Hurwitz encoding shows that the zero dynamics of the Hurwitz flow can store a probability vector exactly, which is a new use of that flow.

For Problem 817, the $k = 4$ case of the exponential rate is settled, the general rates reduce to a certificate problem, and the first steps of the sequence Λ_k are strict. The bridges show how much of the programme is shared structure: several constructions of one workbench are theorems or tools in another.

0.4 How this record was made

This is an audit, carried out by an AI system, of research that was itself carried out with AI systems. The workbenches credit the literature they build on (Terras, Everett and Tao for the cylinder mechanism; Eliahou for the cycle argument; Erdős and Sárközy, Korsky and Costa for Problem 817), and the Problem 817 workbench records that its $k = 4$ proof is due to a contributor on the subreddit r/LLMmathematics whose account was later deleted, reconstructed, checked and partly formalized in Lean by the Clankers, with the finite upper bound formalized by sneed-and-feed. The workbench makes no priority claim for that theorem.

The audit read the proofs, re-derived them, recomputed every finite statement with independent programs, and added the results marked *proved here*, *generalized here* or *computed for this audit*. Statements carry one of four statuses, as in the companion papers: *verified* (proved here or re-computed by a program listed in §6), *not verified here*, *my assessment* (a view with its reasons, in the first person), and *proved negative* (with the counterexample or derivation in the text).

0.5 Organization

Part A contains the Collatz results (§1) and the Problem 817 results (§2). Part B (§3) contains the proved limits of specific methods. Part C (§4) describes the bridges between the workbenches. §5 lists questions and §6 the verification.

Part A. Verified results

1. The Collatz workbench

Notation. X is the set of positive odd integers and $T(n) = (3n + 1)/2^{v_2(3n+1)}$ on X . The exponent word of n of length m is $(a_1, \dots, a_m)$ with $a_i = v_2(3x_{i-1} + 1)$, $x_0 = n$, $x_i = T(x_{i-1})$; $A = a_1 + \dots + a_m$.

1.1 History laws

Proposition 1.1 (cylinders). For every exponent word w of length m and sum A there is a residue h_w modulo 2^A such that an odd $n = 2k + 1$ has exponent word w exactly when $k \equiv h_w \pmod{2^A}$. Among the N consecutive odd starts $2k + 1$, $b \leq k \leq b + N - 1$, the number with word w is $\lfloor N/2^A \rfloor$ or $\lceil N/2^A \rceil$, so the empirical frequency $P(w)$ satisfies $|P(w) - 2^{-A}| < 1/N$.

This is classical (Terras [Te], Everett [Ev]); the workbench credits the mechanism to Tao's Proposition 1.9 [Ta] and claims no priority.

Theorem 1.2 (the full-history threshold; proved here). Let P be the empirical law of the first m exponents over the N starts of Proposition 1.1, $G_m(w) = 2^{-A(w)}$ the model of independent exponents with $\Pr(a = j) = 2^{-j}$, $L = \log_2 N$, $\Delta = TV(P, G_m)$, and Φ the standard normal distribution function.

1. For every $H \geq 0$, $\max\{0, Q_m(H) - N2^{-H-1}\} \leq \Delta \leq \min\{1, Q_m(H) + C(H, m)/N\}$, where $Q_m(H) = \Pr(\text{Bin}(H, 1/2) \leq m - 1)$ and $C(H, m)$ is the binomial coefficient.
2. If $m = \lfloor L/2 + c\sqrt{L} \rfloor$, then $\sup_b |\Delta - \Phi(2c)| \rightarrow 0$ as $N \rightarrow \infty$.
3. If $0 < \delta < 1/2$ and $1 \leq m \leq (1/2 - \delta)L$, then $\Delta \leq N^{-\delta} + N^{-\delta^2/(2 \ln 2)}$. If $m \geq (1/2 + \delta)L$ and $\delta L > 1$, then with $H = \lfloor (1 + \delta)L \rfloor$, $1 - \Delta \leq (1/2)N^{-\delta} + \exp(-2(6L/2 - 1/2)^2/H)$.

All statements are uniform in b .

Proof. Under G_m the sum A is the waiting time for the m -th success in fair coin flips, so $G_m(A > H) = Q_m(H)$, and the words with $A \leq H$ number $C(H, m)$. *Upper bound:* $\Delta = 1 - \sum_w \min(P(w), G(w))$, and by Proposition 1.1 the words with $A \leq H$ contribute at least $1 - Q_m(H) - C(H, m)/N$. *Lower bound:* P is carried by at most N words, each word with $A > H$ has G -mass at most 2^{-H-1} , so $\sum_{A>H} \min(P, G) \leq N2^{-H-1}$, while $\sum_{A \leq H} \min(P, G) \leq 1 - Q_m(H)$. (2) Take $H_+ = L + L^{1/4}$ in the lower bound and $H_- = L - L^{1/4}$ in the upper bound; the error terms tend to 0, and by the central limit theorem, uniform with error $O(H^{-1/2})$, $Q_m(H_{\pm}) \rightarrow \Phi(2c)$. (3) Hoeffding's inequality for $\text{Bin}(H, 1/2)$ in (1), with $H = \lfloor (1 - \delta)L \rfloor$ in the first case and $H = \lfloor (1 + \delta)L \rfloor$ in the second. ■

The workbench's history-law note states the uniformity in b and gives (3) as explicit estimates. Checks: 2,952 sandwich checks; the exact Δ for $N = 2^{10}, 2^{12}, 2^{14}$ at three offsets against (1) for every $0 \leq H \leq 2L + 2$ and against (3); at $N = 2^{20}$ the values 0.004, 0.044, 0.342, 0.668, 0.895 for $c = -1, -1/2, 0, 1/2, 1$ against $\Phi(2c) = 0.023, 0.159, 0.5, 0.841, 0.977$, a slow convergence, as the note says. I have not searched the literature for this threshold.

1.2 Merging families

Lemma 1.3. For even $e \geq 2$, $h_e = (2^e + 2)/3$ is an integer with $h_e \equiv 2 \pmod{4}$ and $v_3(h_e) = v_3(e - 1)$.

Proof. $2^e \equiv 1 \pmod{3}$ gives integrality and $2^e + 2 \equiv 2 \pmod{4}$. With $k = e - 1$ odd, $2^e + 2 = 2(2^k + 1)$, and the lifting-the-exponent lemma gives $v_3(2^k + 1) = 1 + v_3(k)$. ■

Theorem 1.4 (explicit merging families; the workbench's all-even join extension, Theorem 2).

Fix an even $e \geq 2$, $b \geq 1$ and $\sigma \in \{0, 1\}$. Let a be the least $a \geq 1$ with $2^{a+e+2b-\sigma} < 3^{a+b}$, and put $J = 2^{e+2b-\sigma}$, $Q = 3^{a+b}$, $K = 2^a J$, and $(d, r) = (4, 3)$ if $\sigma = 0$, $(32, 27)$ if $\sigma = 1$. Let $n_0 \in (0, dQ)$ solve $n_0 \equiv r \pmod{d}$ and $Jn_0 + h_e 3^b \equiv 0 \pmod{Q}$, and $m_0 = (Kn_0 + 2^{ah_e} 3^b)/Q - 1$. For every $v \geq 0$, $n = n_0 + dQv$ and $m = m_0 + dKv$ are odd, $0 < m < n$, $v_3(n) = b + v_3(e - 1)$, the exponent words of n and m begin with (1) and $(1^a, e, 2^{b-1}, 3)$ if $\sigma = 0$, and with $(1, 2, 1)$ and $(1^a, e, 2^{b-1}, 1, 1, 3)$ if $\sigma = 1$, and the trajectories meet at the same odd number.

Proof. Range of a. At $a = b + 2e - 2\sigma$, $K/Q = (8/9)^{b+e-\sigma} < 1$, and $K < Q$ forces $a \geq e > v_3(e - 1)$. *The 3-adic part.* J is a unit mod 3, so $v_3(n) = b + v_3(e - 1)$; with $z = n/3^b$ and $W = (Jz + h_e)/3^a$ one has $v_2(W) = 1$ and $m = 2^a W - 1$. *The right path.* The first a steps from m have exponent 1 and end at $x_a = Jz + h_e - 1$, odd; by $3h_e - 2 = 2^e$ the next exponent is exactly e , the next $b - 1$ have exponent 2, and the path ends at $4n + 1$ ($\sigma = 0$) or $2n + 1$ ($\sigma = 1$). *The meeting point.* If $\sigma = 0$, $3(4n + 1) + 1 = 4(3n + 1)$ gives the same value as $T(n) = (3n + 1)/2$. If $\sigma = 1$, $n \equiv 27 \pmod{32}$ gives $n \rightarrow (3n + 1)/2 \rightarrow (9n + 5)/8 \rightarrow (27n + 23)/16$ with exponents 1, 2, 1, and $2n + 1 \rightarrow 3n + 2 \rightarrow (9n + 7)/2 \rightarrow (27n + 23)/16$ with exponents 1, 1, 3. *Height.* $m = (K/Q)n + h_e(2/3)^a - 1$ with $K < Q$ and $h_e(2/3)^a < 1$. ■

For example ($e = 8, b = 2, \sigma = 0, a = 16$), $n = 20,241,207 + 1,549,681,956v$ and $m = 14,024,703 + 2^{30}v$ reach $30,361,811 + 2,324,522,934v$ for every $v \geq 0$. The workbench also proves its converse (the endpoint equality forces this progression) and a finite search that decides whether an odd $n > 1$ is the larger member of such a pair; only 367 of the odd $n < 400,000$ admit a template.

1.3 Descent, and a contraction in the fair-coin model

Proposition 1.5 (generalized here). Let $F_m(x) = (3^m x + C_m)/2^{\{A_m\}}$ be the affine map of the first m odd steps of an exponent word, and n_0 the least odd $n \geq 3$ whose word begins with it. Then every odd $n \geq 3$ whose word begins with it satisfies $T^m(n) < n$ if and only if $F_m(n_0) < n_0$; this holds in particular if $F_m(3) < 3$. More precisely, a member descends if and only if $D = 2^{\{A_m\}} - 3^m > 0$ and $n > C_m/D$.

Proof. $F_m(x) - x = (C - Dx)/2^{\{A_m\}}$ with $C = C_m > 0$, and $T^m(n) = F_m(n)$ for every member. ■

The workbench's Chapter 2, Theorem 2.1, assumes $F_j(3) \geq 3$ for $j < m$ and $F_m(3) < 3$; the criterion needs only $F_m(n_0) < n_0$, which is also necessary. For example, the word $(1, 1, 1, 1, 4)$ has $F_5(3) = 235/64 > 3$, but $n_0 = 95$ and every member descends.

Proposition 1.6 (generalized here). For $x \geq 2$, $(1/2)(\sqrt{(x-2)/2} + \sqrt{(3x-1)/2}) \leq (\sqrt{15/4})\sqrt{x-1}$, with equality exactly at $x = 7$.

Proof. With $u = (x-2)/2$ and $v = (3x-1)/2$, Cauchy-Schwarz gives $\sqrt{u} + \sqrt{v} \leq \sqrt{(2u+v) \cdot (3/2)}$, and $2u+v = (5/2)(x-1)$; equality needs $4u = v$. ■

In the fair-coin model, where x moves to $x/2$ or $(3x+1)/2$ with probability $1/2$ each, this says $E\sqrt{X_{\text{next}} - 1} \leq (\sqrt{15/4})\sqrt{x-1}$. Started at $X_0 = 3$ with the first step odd, the first time τ below 3 satisfies $\Pr(\tau > h) \leq (3/2)(31/32)^h$, and the completed stopping word has tail $t_H \leq (31/20)(31/32)^H$ (the workbench's (7.9)–(7.16)). These are statements about the model, as the workbench says.

1.4 Cycles

Proposition 1.7. No nontrivial positive cycle of T has at most one step with exponent 1.

Proof. Let $n > 1$ be the least element. A step with exponent ≥ 2 from $x > 1$ decreases, so the step from n has exponent 1 and every later step decreases; an exponent ≥ 3 later would give a value below n . So the word is $(1, 2, \dots, 2)$, and closing the cycle gives $nD = 2 \cdot 4^{\{m-1\}} - 3^{\{m-1\}}$ with $D = 2 \cdot 4^{\{m-1\}} - 3^m$ odd and prime to 3, so $D = \pm 1$; but $D = -1$ for $m = 1, 2$ (the negative cycles -1 and -5) and $D_{\{m+1\}} = 4D_m + 3^m \geq 5$ for $m \geq 2$. ■

Theorem 1.8. Every nontrivial positive cycle of T has at least 1636 odd steps, at least 2593 halvings, and more than 678 steps with exponent 1.

Proof. The workbench verifies that every odd $n \leq 330,749$ reaches 1 (recomputed here). Let the cycle have m odd steps, k of them with exponent 1, total exponent A and least element $s \geq 330,751$. Multiplying $T(x_i)/x_i = (3 + 1/x_i)/2^{\{a_i\}}$ around the cycle gives $2^A \in (3^m, (3 + 1/s)^m]$, and $A \geq 2m - k$ gives $(4s)^m \leq 2^k(3s+1)^m$. The least m for which a power of 2 lies in that interval is $m = 1636$ (exact integer arithmetic), so $A > 1636 \log_2 3 = 2592.998\dots$, and at $m = 1636$ the inequality needs $k \geq 679$. ■

This is Eliahou's argument [El]; the count k differs from the number of local minima used in the published bounds of Simons-de Weger [SdW] and Hercher [He]. Run with a larger verification range it gives more. A verification made for this record shows that every odd n with $3 \leq n < 2^{33}$ falls below itself (38 seconds on two cores, 128-bit arithmetic, largest value met $9.07 \cdot 10^{18}$; repeated by two independently written programs), so every $n < 2^{33}$ reaches 1, and the same argument gives $m \geq 190,537$, $A \geq 301,994$ and $k \geq 79,080$.

Corollary 1.9 (from Barina's range). Every $n < 2.17 \cdot 10^{20}$ reaches 1: Barina verified every $n < 2^{68}$ (2020, published 2021 [Ba1]) and every $n < 2^{71}$ [Ba2]. Hence every nontrivial positive cycle has at least 72,057,431,991 odd steps, at least 114,208,327,604 halvings, and at least 29,906,536,378 steps with exponent 1.

Proof. The least element s exceeds $s_{\text{lo}} = 1/(2^{\{10439860591/6586818670\}} - 3) = 2.1689015534 \cdot 10^{20}$. So $A/m \in (\log_2 3, \log_2(3 + 1/s_{\text{lo}}))$, an interval that excludes $10439860591/6586818670$, and every fraction in $(\log_2 3, 10439860591/6586818670)$ has denominator at least 72,057,431,991, the denominator of the next best upper approximation of $\log_2 3$. The bounds on A and k follow as in Theorem 1.8. ■

The fraction $114208327604/72057431991$ was found by the continued-fraction algorithm for the simplest fraction in an interval (80-digit arithmetic) and confirmed by a Stern–Brocot descent (120-digit arithmetic); the same computation returns Eliahou’s published $A = 17,087,915$ at $s = 2^{40}$. The published bound is stronger: Hercher [He, Corollary 29] proves that if every $n \leq 3 \cdot 2^{69}$ reaches 1, every nontrivial cycle contains more than $1.375 \cdot 10^{11}$ odd numbers, and Barina’s verification passed that range in 2023 [Ba2]. Hercher’s proof is not verified here.

1.5 Restatements, and the Tao-clock deduction

The workbench states two exact restatements of the conjecture: the vanishing of $\ker(H^1(K_\varepsilon) \rightarrow H^1(K))$ for the orbit-graph complex K and its first-jet thickening, a group isomorphic to the sum of $\mathbb{Z}/m_C$ over the nontrivial cycles and of $\mathbb{Z}$ over the components without a cycle (Chapter 9; not verified here beyond its statement), and the vanishing of H_0 of a relative orbit-graph homology (Chapter 3; verified). Its Tao-clock preprint deduces from Tao’s Proposition 1.11 [Ta] that the laws of the first entry of an orbit into $[1, x]$, started from the logarithmic law on $[t_{\{j+1\}}, t_{\{j+1\}}^\alpha]$ with $t_j = b^\alpha$, $\alpha = 1001/1000$, converge in ℓ^1 as $j \rightarrow \infty$ at the rate $L_c(\log t_j)^{-c}$, coherently in x , with joint laws for several thresholds converging with the same error. The deduction (nested first passage, ℓ^1 contraction, a geometric sum of Tao’s adjacent-scale errors, completeness) is verified; the preprint’s repairs to Tao’s §§5–6 are not verified here.

1.6 The Hurwitz zero-cluster encoding

Fix (m, A) . The K odd-return words, the exponent words $w = (a_1, \dots, a_m)$ with all $a_i \geq 1$ and sum A , have pairwise distinct residues r_w modulo 2^{A+1} . For a probability vector λ on these words put $Z_\lambda(s, t) = \sum_w \lambda_w \zeta(s, 1 + tr_w)$ for $|t|(2^{A+1} - 1) < 1$.

Theorem 1.10 (proved here for simple zeros; verified in general). Let ρ be a nontrivial zero of ζ . The t -jet of order J of the zeros of $Z_\lambda(\cdot, t)$ near ρ (of their monic cluster polynomial if ρ is multiple) determines the moments $M_i = \sum_w \lambda_w r_w^i$, $i \leq J$, and conversely. Hence the jet of order $K - 1$ determines λ , and for $K \geq 2$ the jet of order $K - 2$ determines λ at no interior point of the simplex.

Proof for a simple zero. $\zeta(s, 1 + u) = \sum_j b_j(s)u^j$ with $b_j(s) = (-1)^j(s)_j \zeta(s + j)/j!$ [DLMF, 25.11.10], so $Z_\lambda = \sum_j b_j(s)M_j t^j$ with $M_0 = 1$, and $b_j(\rho) \neq 0$ for $j \geq 1$. Writing the zero as $\rho(t) = \rho + c_1 t + c_2 t^2 + \dots$, the coefficient of t^J in $Z_\lambda(\rho(t), t) = 0$ is $\zeta'(\rho)c_J + b_J(\rho)M_J$ plus a polynomial in the lower c_i and M_i , so the c ’s and M ’s determine each other triangularly. The K distinct nodes r_w make $M_0, \dots, M_{K-1}$ determine λ (Vandermonde), and with $h_w = 1/[\prod_{v \neq w} (r_w - r_v)]$, $\sum_w h_w r_w^i = 0$ for $i \leq K - 2$, so $\lambda + \varepsilon h$ has the same moments to order $K - 2$. ■

The encoding is an exact, invertible embedding of weights on residue classes into the motion of a zeta zero under the Hurwitz flow $t \mapsto \zeta(s, 1 + t)$. It works at every nontrivial zero, so it is a storage device for the weights rather than a constraint on where the zeros lie; the note says so. The same note gives three witnesses of what coarser encodings lose: two parity words of length 5 whose affine constant terms $1/2 + q/32$ and $1/8 + q/16 + q^2/32$ differ by $(q - 3)(q + 4)/32$ and agree at $q = 3$; the eigenvalues of the affine matrix forget bit order; and $(K - 2)$ -jets do not separate weights.

2. The Erdős Problem 817 workbench

For a finite set A of distinct positive integers let $H(A) = \{\sum_{a \in A} \varepsilon_a a : \varepsilon_a \in \{0, 1\}\}$ and $S(A) = \sum_{a \in A} a$. A is k -admissible if $H(A)$ contains no k -term arithmetic progression with nonzero step, and $g_k(n)$ is the least N such that some n -element $A \subseteq \{1, \dots, N\}$ is k -admissible. Erdős Problem 817, a problem of Erdős and Sárközy, asks for estimates of $g_k(n)$, in particular whether $g_3(n) \gg 3^n$; they proved $g_3(n) \gg 3^n/n^{O(1)}$ [ES]. The workbench reports that Costa’s preprint [Co] proves $\liminf g_3(n)/3^n = 0$, a

negative answer to that question; this is not verified here, and the workbench's own results concern $k \geq 4$. Admissibility is hereditary, so g_k is strictly increasing.

2.1 The rate $19^{\{1/3\}}$ for four-term progressions

Theorem 2.1. For every $n \geq 1$, $(19^{\{n/3\}} - 1)/(2n) \leq g_4(n) \leq 8 \cdot 19^{\{[n/3]-1\}}$. Hence $\lim_n g_4(n)^{\{1/n\}} = 19^{\{1/3\}} \approx 2.6684$.

The workbench's paper is complete and correct as written, and the proof below follows it. The workbench reports the upper half as Lean-checked; the Lean files were read, not built. Let $A = \{a_1, \dots, a_n\}$ be 4-admissible. A relation is $c \in \{-2, \dots, 2\}^n$, $c \neq 0$, with $\sum c_{i+j} = 0$.

Lemma 2.2 (relation splitting). If c is a relation, then with $P_j = \sum \{c_i = j\}$ a_i and $N_j = \sum \{c_i = -j\}$ a_i one has $P_1 = N_1$ and $P_2 = N_2$.

Proof. $X_0 = P_1 + N_2$, $X_1 = P_1 + P_2$, $X_2 = N_1 + N_2$, $X_3 = N_1 + P_2$ are subset sums. With $d = P_2 - N_2$, the relation gives $N_1 - P_1 = 2d$, and then $X_1 - X_0 = X_2 - X_1 = X_3 - X_2 = d$. Admissibility forces $d = 0$. ■

Lemma 2.3 (blocks). Call a ± 1 -relation minimal if its support is inclusion-minimal. Two minimal relations have disjoint supports or are equal up to sign, and every relation is $\sum_j t_j r^{\{j\}}$ with $t_j \in \{-2, \dots, 2\}$, where the blocks $r^{\{j\}}$ are the minimal relations, with pairwise disjoint supports of size at least 3.

Proof. For minimal r, s , the ± 2 layers of $r \pm s$ are $r \cdot 1_{\{E \pm\}}$ with $E_{\pm} = \{i : r_i = \pm s_i \neq 0\}$; by Lemma 2.2 each is a relation or zero, and minimality gives the dichotomy. Every ± 1 -relation is a sum of blocks with disjoint supports, by induction on the support; for a relation c , the ± 1 part and half the ± 2 part are ± 1 -relations with disjoint supports (Lemma 2.2). ■

Lemma 2.4 (the ternary count). Let $T(A) = \{\sum x_i a_i : x \in \{0, 1, 2\}^n\}$, let the blocks have sizes $m_1, \dots, m_h$, and let $t = n - \sum m_j$. Then $|T(A)| = 3^t \prod_j (3^{m_j} - 2^{m_j})$.

Proof. Two vectors have the same value iff their difference is $\sum t_j r^{\{j\}}$; on a block, reflecting $x_i \mapsto 2 - x_i$ where $r_i = -1$ turns the equivalence into translation by $(1, \dots, 1)$, and each class has exactly one representative with minimum 0; these number $3^m - 2^m$. ■

Proof of Theorem 2.1. Lower bound: $f_m = 3^m - 2^m$ has $f_3 = 19$ and $f_{m+1} = 3f_m + 2^m > 3f_m$, so $f_m \geq 19^{\{m/3\}}$ for $m \geq 3$, and $3 \geq 19^{\{1/3\}}$; so $|T(A)| \geq 19^{\{n/3\}}$, and $T(A) \subseteq [0, 2S(A)] \subseteq [0, 2nN]$. Upper bound: $D = H(\{1, 7, 8\}) = \{0, 1, 7, 8, 9, 15, 16\}$ contains no nonconstant 4-term progression modulo 19 (all 19·18 checked). For $A_m = \{19^j w : j < m, w \in \{1, 7, 8\}\}$, $H(A_m)$ is the set of integers whose base-19 digits lie in D , without carries. If $x + id$ ($0 \leq i \leq 3$) lie in $H(A_m)$ with $d = 19^r e$, $19 \nmid e$, the r -th digits form a nonconstant progression in D modulo 19. So A_m is 4-admissible with $3m$ elements and maximum $8 \cdot 19^{\{m-1\}}$. ■

Proposition 2.5 (lifting; generalized here). Let each W_j be $\{1, 7, 8\}$ or $\{2, 3, 5\}$. For every 4-admissible B and $q \geq 0$, $\{19^j w : j < q, w \in W_j\} \cup 19^q \cdot B$ is 4-admissible. Hence $g_4(n + 3q) \leq 19^q g_4(n)$; $g_4(n) \leq c \cdot 19^{\{[n/3]-1\}}$ with $c = 1, 3, 5$ for $n \equiv 1, 2, 0 \pmod{3}$; $g_4(3q) \leq 79 \cdot 19^{\{q-2\}}$ and $g_4(3q + 1) \leq 246 \cdot 19^{\{q-2\}}$ for $q \geq 2$, and $g_4(3q + 2) \leq 40 \cdot 19^{\{q-1\}}$ for $q \geq 1$, that is at most 0.219, 0.256 and 0.296 times $19^{\{n/3\}}$; and for every certificate (q, B) of Theorem 2.7, $g_k(m|B| + n) \leq q^m g_k(n)$. The sets $\{1, 7, 8\}$ and $\{2, 3, 5\}$ are the only 3-element B with $S(B) < 19$ and $H(B)$ free of nonconstant 4-term progressions modulo 19.

Proof. H of the lifted set is $H(A) + 19^q \cdot H(B)$ without carries. A progression with $19^q | d$ reduces to one in $H(B)$; otherwise the r -th digits give a nonconstant progression modulo 19 in $H(W_r)$, and $H(\{2, 3, 5\}) = 3 \cdot D \pmod{19}$. The last statement is a finite search. ■

Theorem 2.6 (the variance refinement). Write $n = 3q + r$ ($0 \leq r \leq 2$) and $M_n = 19^q 3^{\{r\}}$. Every 4-admissible A has $|T(A)| \geq M_n$ and $19(|T(A)|^2 - 1) \leq 192 \sum a_i^2$. Hence $g_4(n) \geq \sqrt{(19(M_n^2 - 1)/(192n))}$, about $c_r 19^{\{n/3\}}/\sqrt{n}$ with $c_r = 0.315, 0.354, 0.398$, and $g_4(n) \geq [(M_n + n(n-1) - 1)/(2n)]$.

Proof. $f_m \geq 19^{\{[m/3]\}} 3^{\{m \bmod 3\}}$, and since $3^3 > 19$ the least value of the product in Lemma 2.4 is M_n . The uniform law on $T(A)$ is the law of a sum of independent components; a block with coefficients b_i , $\sum b_i = 0$, has variance $\kappa_m Q$ with $\kappa_m = (8 \cdot 3^m - 3 \cdot 2^m)/(12(3^m - 2^m)) \leq 16/19$ for $m \geq 3$, and a

free coordinate has variance $(2/3)a_i^2$. So the variance is at most $(16/19)\sum a_i^2$, while M distinct integers have uniform variance at least $(M^2 - 1)/12$. ■

2.2 General k

Theorem 2.7. For every $k \geq 3$, $\Lambda_k = \lim_n g_k(n)^{1/n}$ exists and equals $\inf q^{1/|B|}$ over the pairs (q, B) with $S(B) < q$ and $H(B)$ modulo q free of k -term progressions with nonzero step. Moreover $\Lambda_3 = 3 > \Lambda_4 = 19^{1/3} > 97^{1/6} \geq \Lambda_5$ (proved here).

Proof. If A and B are k -admissible and $R = 2S(A) + 1$, then $C = A \cup R \cdot B$ is k -admissible and $2S(C) + 1 = (2S(A) + 1)(2S(B) + 1)$, since a progression in $H(C) = H(A) + R \cdot H(B)$ has second differences $\Delta^2\alpha + R\Delta^2\beta = 0$ with $|\Delta^2\alpha| < R$. So $F_k(n) = \min(2S(A) + 1)$ is submultiplicative, and Fekete's lemma with $2g_k(n) + 1 \leq F_k(n) \leq 2ng_k(n) + 1$ gives the limit. A certificate (q, B) gives k -admissible sets $\{q^j b\}$ as in Theorem 2.1, and every k -admissible B gives the certificate $(2S(B) + 1, B)$. For $k = 3$: the certificate $(3, \{1\})$ gives $\Lambda_3 \leq 3$; conversely a 3-admissible set has no relation with coefficients in $[-2, 2]$ (X_0, X_1, X_2 would be a 3-term progression), so the 3^n ternary sums are distinct and $3^n \leq 2nN + 1$. ■

The workbench also proves $\Lambda_k \geq (k - 1)/(k - 2)$, the monotonicity $\Lambda_{k+1} \leq \Lambda_k$, and that avoidance of k -term progressions by base- q numbers with digits from a finite set is decided by a finite carry graph.

Theorem 2.8 (certificates for $k = 5, 6$).

1. $B^* = \{1, 4, 5, 17, 21, 22\}$ has $S(B^*) = 70$, $|H(B^*)| = 38$, and $H(B^*)$ is free of nonzero-step 5-term progressions modulo 97 and of 6-term progressions modulo 93. Hence $\Lambda_5 \leq 97^{1/6} \approx 2.1435$ and $\Lambda_6 \leq 93^{1/6} \approx 2.1285$.
2. $A^\# = \{3, 4, 7, 34, 37, 41, 216, 250, 253, 257\}$, the ten pairwise distances of the marks 0, 4, 7, 41, 257, has $S(A^\#) = 1102 < 1651 = 13 \cdot 127$, $|H(A^\#)| = 291$, and $H(A^\#)$ is free of nonzero-step 6-term progressions modulo 1651. Hence $g_6(n) \leq 257 \cdot 1651^{\lfloor n/10 \rfloor - 1}$ and $\Lambda_6 \leq 1651^{1/10} \approx 2.0979$.

Each modular condition was recomputed twice. Korsky's preprint [Ko] proves, for fixed $k \geq 4$, $g_k(n) \gg_k ((k - 1)/(k - 2))^n n^{-\log_2((k - 1)/(k - 2))}$ and upper rates $\min_p p^{2/(\min\{p, k\} - 1)}$ over primes $p \geq 3$: $5^{2/3} \approx 2.924$ for $k = 4$, $\sqrt{5} \approx 2.236$ for $k = 5$ and $7^{2/5} \approx 2.178$ for $k = 6$. Theorem 2.1 determines the $k = 4$ rate between Korsky's $3/2$ and $5^{2/3}$, and Theorem 2.8 lowers the upper rates for $k = 5$ and 6.

Proposition 2.9 (distance rulers; generalized here). If A contains r pairwise disjoint subsets each summing to L , then $H(A)$ contains $0, L, \dots, rL$. In particular, for marks $0 = t_0 < \dots < t_{m-1} = L$ whose $2m - 3$ distances $L, t_i, L - t_i$ are pairwise distinct, the subset sums of any set containing these distances contain $0, L, \dots, (m - 1)L$. For $A^\#$ this gives $0, 257, 514, 771, 1028$, so $A^\#$ is 6-admissible and not 5-admissible.

2.3 Exact small values

Proposition 2.10 (computed for this audit). $g_4(1), \dots, g_4(6) = 1, 3, 5, 14, 40, 79$, and $176 \leq g_4(7) \leq 246$.

The values for $n \leq 6$ were computed by four independent exhaustive searches, which agree, including the numbers of optimal sets 1, 2, 2, 2, 7, 1. The unique optimal 6-set is $\{2, 29, 45, 74, 77, 79\}$: two blocks $(29 + 45 = 74, 2 + 77 = 79)$, and $|T| = 361 = M_6$, the least value Theorem 2.6 allows. $g_4(7) \leq 246$ by $\{1, 3, 39, 180, 219, 243, 246\}$; the lower bound 176 is conditional on the search program, which excluded maxima in [132, 175] by pruning from Lemma 2.4 and Theorem 2.6. I have not found these values in the sources read; OEIS could not be consulted.

2.4 The continuation notes

The workbench's 29 continuation notes are ordinary proofs with exact finite certificates. Beyond the results above: a finite-period approximation of the best lower-limit growth rate of q -ary images (verified in part; the radix-ten example has minima $a_1 = 13$, $a_{3r} = 893^r$, $a_{3r+1} = 93^2 \cdot 893^{r-1}$, $a_{3r+2} = 93 \cdot 893^r$, verified for $m \leq 16$ by an independently built automaton); a correction the workbench published (the independence test $Y_P = P \cup (L - P)$ with $P \subseteq [0, Q]$ needs $L > 3Q$, not $L > 2Q$); and a recurrence control: if $a_j > S_{j-1} + S_{j-r}$ for all j , the subset sums contain no $(2^r + 1)$ -term progression, which for $r = 2$ covers the Pell numbers. The other continuation notes are not verified here.

Part B. Proved limits of specific methods

3. What specific methods cannot do

Collatz.

- *The full-history threshold* (Theorem 1.2(3)): above $(1/2 + \delta)\log_2 N$ halving counts, the full word law of N consecutive starts is far from the geometric model. The statement concerns the full word law; coarser observables and individual orbits are outside it.
- *Coarser encodings* (§1.6): the $(K - 2)$ -jets of the Hurwitz encoding do not separate weights, while the $(K - 1)$ -jets do.
- *The formal inverse* $(I - tP)^{-1}$ of Chapter 3 is not a finite integral chain on cycles or divergent rays, and the signed primitive of the unit cochain is unbounded below on an acyclic component. These identify inputs that a global argument along those lines would need.
- *The finite template search* of Chapter 20 decides the template property at each n ; it does not by itself show that a template applies at every retained source.

Problem 817.

1. The count of 19 classes per 3-block needs admissibility: $\{1, 2, 3\}$ has $|T| = 13 < 19$.
2. Admissible classes are strictly nested: $\{1, 2\}$ is 5-admissible but not 4-admissible.
3. Modular certificates are sufficient, not necessary, for all-length admissibility ($B = \{2, 7\}$, $q = 11$, $k = 3$); Theorem 2.7 is unaffected.
4. For B^* , every base 71-96 fails at $k = 5$ and every base 71-92 at $k = 6$.
5. Every distinct-distance ruler with m marks has an m -term progression (Proposition 2.9), and the ruler $(0, 1, 5, 22)$ has no 6-admissible extension by one further mark with distinct distances.
6. For $A^\#$ no canonical schedule with radix in $[1103, 1650]$ is admissible.
7. Image counts do not determine admissibility.

Each concerns the named block or construction; the unrestricted rates Λ_k for $k \geq 5$ are open.

Part C. Bridges between the workbenches

4. The bridges, what crosses them, and what they suggest

The workbenches were built as a connected programme, and many of their constructions cross from one workbench to another. This section states each crossing as a typed map: what it carries, what is established, and what it suggests. The types are P (a proved map or lemma used on both sides), I (the same identity or object on both sides), D (one side imports the other's theorem), R (a regression test), and A (an analogy or proposal).

4.1 The graph

The nodes are the zeta programme (zeta), Erdős–Straus (ES), Collatz (CZ), Erdős 817 (EP), Yang–Mills (YM), Navier–Stokes (NS), S^6 , and the Jacobian map F.

#	Edge	What crosses
1	zeta $\rightarrow$ CZ	the Hurwitz zero-cluster reconstruction, re-proved in the part used (§1.6)
1a	zeta $\rightarrow$ CZ	the split-zero support modules and killed-representative quotient, on the Collatz o
2	NS $\rightarrow$ ES	the torus endomorphism $J = [[3, 1], [1, 5]]$ of the NS manuscript, as an operator
3	$S^6 \rightarrow$ ES	the monodromies A_1, A_2, A_∞ of the $(3, 4, \infty)$ period system
4	$S^6 \rightarrow$ YM	the imaginary period block of the period matrix, through one positive number

#	Edge	What crosses
5	F → NS kinematics → YM	incompressible flows, the escaping curve, the material tensor, trial states
6	F → zeta	an inverse-fibre chart, fibre heat, a material generator, endpoint jets
7	ES ↔ zeta	a four-dimensional Keller map P with $\det DP = -2$ and its conductor
8	zeta → YM	the Gram sandwich from a relative column error, $\delta \leq 1$
9-10	zeta → YM	an inverse-power observability pattern; a composed minimum-lift identity
11	NS → YM	velocity ↦ reducible SU(2) connection, $-2\sum \text{tr } F_{ij}^2 = \lambda^2$
12	NS → zeta	the NS field as arithmetic input (Mellin transports)
13	S ⁶ , F → NS	fluid coordinates; S ⁶ gauge-scalar equations → NS on T ³
14-15	zeta ↔ ES	zeta results as dependencies of ES items; the ES reader as a source dependency
16	EP → zeta	finite word sources in the supported quotient; a cocycle identity
17	ES ↔ CZ	$3(4n + 1) + 1 = 4(3n + 1)$
18	CZ ↔ ES ↔ zeta	the homogenization $(x, h) \mapsto (ax + bh, h)$
19	CZ → YM	$X_v - X_u = (q - 3)(q + 4)/32$
20	ES ↔ zeta	the semiring $G(R) = R \sqcup \{\tau\}$
21	ES ↔ zeta	a finite Weyl basis of $\text{End } \mathbb{C}[A]$
22	ES ↔ S ⁶	the Niemeier lattice with root system A ₅ ⁴ D ₄ , glue index 72
23-28	various	shared vocabulary, routing proposals, packages

4.2 The substantive bridges

zeta → Collatz: zero motion as storage (edge 1). The map from probability vectors λ to the order-J jet of the zero cluster of $\sum_w \lambda_w \zeta(s, 1 + \text{tr}_w)$ near a nontrivial zero ρ factors as a moment map $\lambda \mapsto (M_1, \dots, M_J)$, depending only on the residues, followed by a triangular polynomial bijection whose diagonal coefficients are $\zeta'(\rho)$ and $b_J(\rho)$ (Theorem 1.10). The zeta programme studies the same flow $t \mapsto \zeta(s, 1 + t)$ at $t = 0$ through its jets; here the motion of one zero, taken at several shifts and averaged with weights, stores the weights exactly. The Collatz workbench is the only one besides the zeta programme that uses this flow. My assessment: the bridge shows that the Hurwitz flow is a faithful carrier of finite probability data, which suggests using zero dynamics as a coordinate system for other finite laws, and conversely using known laws to probe the dynamics of zeros.

NS → ES: an operator from fluid dynamics in shell arithmetic (edge 2). The torus endomorphism J of the NS manuscript (determinant 14, eigenvalues $4 \pm \sqrt{2}$) enters 21 Erdős–Straus statements. Their content is the identity $\{(u, v) \in G^2 : u^3v = 1 = uv^5\} = \{(u, u^{-3}) : u^{14} = 1\}$ in a finite abelian group G (in the application the unit group modulo R_a), which gives exact aliasing on finite character groups, and the sharp constant $\inf_{\{k \neq 0\}} \|v_k\| \|k\| = 1/\sqrt{4 + 2\sqrt{2}}$ for the eigen-directions $v_r = (1, 1 - \sqrt{2})$ and $v_t = (\sqrt{2} - 1, 1)$, approached along Pell vectors. The bridge carries the operator and its Diophantine approximation properties; the NS existence theorem is not used.

S⁶ → ES and S⁶ → YM: two uses of one period system (edges 3, 4). Erdős–Straus uses the integral monodromies of the $(3, 4, \infty)$ period system to build covers whose sections are the solutions (ES record, §10); Yang–Mills uses the imaginary period block as a magnetic background, through one positive number D. Both use the displayed period matrices, and the S⁶ monodromy record determines the group that governs the first use.

The name Fable: three objects (edges 5-7). In the repositories the name refers to the three-dimensional Keller map F of the Jacobian counterexample, to the S⁶ construction in older files, and to the four-dimensional Keller map P of the Erdős–Straus crosswalk. F enters Yang–Mills through incompressible flows and trial states that track the free two-gluon threshold (YM record, §5); P has $\det DP = -2$, and its fibres, monodromy and conductor are not verified here.

zeta → YM: linear algebra shared across programmes (edges 8-10). What crosses is the Gram sandwich $(1 - \delta)^2 \Phi^* \Phi \leq \Phi_J^* \Phi_J \leq (1 + \delta)^2 \Phi^* \Phi$ when $\|(\Phi_J - \Phi)x\| \leq \delta \|\Phi x\|$ and $\delta \leq 1$ (the zeta instance takes δ from a dilation factor; the Yang–Mills instance from a heat semigroup), an observability pattern, and Schur-complement identities. The transfers state that they assert no map between arithmetic zeta objects and gauge fields.

NS → YM (edge 11). A divergence-free velocity u becomes the reducible connection $A_i = \lambda u_i T$ with curvature density $\lambda^2 |\text{curl } u|^2$. The resulting curvature rates are derived, conditionally on four statements of OpenAI’s manuscript, in the NS record (Theorem 3.1); the zero-quotient diagonal exists with coupling tending to 0.

Erdős 817 → zeta (edge 16). Ten interface notes restate 817’s finite constructions in the zeta programme’s supported-quotient vocabulary. The one verified is a correct restatement: free vector spaces on coefficient words, the evaluation quotient, the cut map $e_{\{(x,y)\}} \mapsto e_{\{x+Q(u)y\}}$ with $\dim \ker = F(u)F(v) - F(uv)$, and the cocycle identity $\kappa(u, v)\kappa(uv, w) = \kappa(v, w)\kappa(u, vw)$ for $\kappa(u, v) = F(u)F(v)/F(uv)$.

4.3 Shared identities

- **ES and Collatz** (edge 17). $3(4n + 1) + 1 = 4(3n + 1)$, so n and $4n + 1$ have the same odd successor. It is the Erdős–Straus workbench result R05 (which credits the underlying diagrams to u/CivQ17) and the meeting-point step of Theorem 1.4.
- **Collatz, ES and zeta** (edge 18). The homogenization $(x, 1) \mapsto (x, h)$, $(x, h) \mapsto (ax + bh, h)$ makes affine branch maps commute with support inclusions; the zeta foundation cites it from the Clankers’ Zenodo record 19900461.
- **Collatz and YM** (edge 19). The constant terms $1/2 + q/32$ and $1/8 + q/16 + q^2/32$ of two parity words differ by $(q - 3)(q + 4)/32$ and agree at $q = 3$.
- **ES and zeta** (edges 20, 21). In $G(R) = R \sqcup \{\tau\}$, $\{\tau\}$ is a prime ideal and $\{\tau, 0_R\}$ is prime iff R is a domain; the finite Weyl operators T_{aM_χ} form an orthogonal basis of $\text{End } C[A]$.
- **ES and S⁶** (edge 22). Both name the Niemeier lattice with root system $A_5^4 D_4$; $|\text{disc}(A_5^4 D_4)| = 6^4 \cdot 4 = 5184$, so the glue index is 72. The Erdős–Straus glue code (order 72, isotropic, cost spectrum $\{0: 1, 4: 46, 6: 25\}$) was checked exhaustively.

4.4 What the bridges suggest

Several bridges are exact restatements: the shell criterion as sections of covers, the Collatz conjecture as the vanishing of an abelian group, the Hurwitz encoding, and 817’s constructions in the supported-quotient vocabulary. A restatement moves a problem into a new language, and it becomes a tool when the new side supplies a constraint or a technique that the old side lacks. My assessment of where that is most likely: the S^6 monodromy has already supplied one, the complete orbit classification of the Erdős–Straus covers; the Hurwitz flow supplies an exact storage of finite laws that could be turned into a probe of zero dynamics; and the shared linear algebra between the zeta programme and Yang–Mills (Gram sandwiches, observability) is a common toolkit whose further uses can be tested quickly. For each restatement, identifying the constraint that the new side contributes is the open task (Question 6).

5. Questions

1. **The threshold.** Is the Gaussian transition of Theorem 1.2 in the literature, and what is the analogous threshold for coarser observables, such as the law of A alone?
2. **Merging.** Does every retained source of the workbench’s template search admit a merging template?
3. **Cycles.** What bound does the method of Corollary 1.9 give from the current verification range, and how does it compare with Hercher’s refinement?
4. **Problem 817.** Is $\Lambda_5 < 97^{1/6}$? What are $g_4(7)$ and $g_4(8)$?
5. **Λ_3 and Costa’s preprint.** What is the correct asymptotic of $g_3(n)$?
6. **Bridges.** For each exact restatement of §4.4, what constraint does the new side contribute?

6. Verification

The scripts are in the record’s `checks/collatz_ep817/` folder, with their outputs:

- the Collatz checks of the history law, the merging families (600 + 1,200 pairs), the descent criterion (36,267 cases and all 637 words with $m \leq 5$, $A \leq 10$), the cycle arguments and the Tao-clock identities (`check_history_law_indep.py`, `check_all_even.py`, `check_all_even_large.py`, `check_transport_cycles_indep.py`, `check_tao_clock_indep.py`, `check_split_zero_indep.py`);

- the verification of all odd $n < 2^{33}$ (descent_2e33.c, 38 seconds) and the continued-fraction computations of Corollary 1.9;
- the Problem 817 checks: the certificates of Theorem 2.8, the lifting sets (28 sets), the exhaustive searches of Proposition 2.10, and the continuation checks (ep817_k4_checks.py, ep817_continuation_checks.py, ep817_fp3_matrix_check.py, g4_bitset.c);
- the bridge identities of §4.3 (generality_checks.py, workbench_reader_claims_checks.py).

References

- [Ba1] D. Barina, Convergence verification of the Collatz problem, J. Supercomputing 77 (2021) 2681–2688.
- [Ba2] D. Barina, Improved verification limit for the convergence of the Collatz conjecture, J. Supercomputing 81 (2025), article 810.
- [Co] S. Costa, A negative answer to the Erdős–Sárközy question, arXiv:2609.06303 (2026).
- [DLMF] NIST Digital Library of Mathematical Functions, §25.11.
- [El] S. Eliahou, The $3x+1$ problem: new lower bounds on nontrivial cycle lengths, Discrete Math. 118 (1993) 45–56.
- [ES] P. Erdős, A. Sárközy, Arithmetic progressions in subset sums, Discrete Math. 102 (1992) 249–264.
- [Ev] C. J. Everett, Iteration of the number-theoretic function $f(2n) = n$, $f(2n+1) = 3n+2$, Adv. Math. 25 (1977) 42–45.
- [He] C. Hercher, There are no Collatz m -cycles with $m \leq 91$, J. Integer Seq. 26 (2023), Article 23.3.5.
- [Ko] S. Korsky, Arithmetic progression-free subset-sum sets, arXiv:2606.24139 (2026).
- [SdW] J. Simons, B. de Weger, Theoretical and computational bounds for m -cycles of the $3n+1$ problem, Acta Arith. 117 (2005) 51–70.
- [Ta] T. Tao, Almost all orbits of the Collatz map attain almost bounded values, Forum Math. Pi 10 (2022), e12.
- [Te] R. Terras, A stopping time problem on the positive integers, Acta Arith. 30 (1976) 241–252.